

A possible role for the alpha 1-->3 galactosyl epitope and the natural anti-gal antibody in oncogenesis.



Glycoconjugates and their antibodies are vital components of host-tumor interaction. This review concentrates on the oncological implications of research concerning the alpha gal triad; the alpha 1-->3 galactosyl epitope (alpha Gal), the enzyme responsible for its construction, alpha 1,3 galactosyl transferase (alpha 1-3GT), and its associated antibody: anti-gal. Alpha gal epitopes, previously assumed to be absent from human tissue, have been demonstrated on several human cancer cell lines, senescent red blood cells, and Graves' disease thyrocytes. Alpha-gal presence on neoplastic lines is correlated with increased metastatic formation in animal models. The mechanisms of human response to these neoantigens are complex, as natural anti-gal antibodies exist in high titers in normal sera, thus predicting immunological recognition of cells expressing alpha gal epitopes. Hypotheses vary regarding the pathogenic contributions of metastasis-associated phenomena such as de novo expression of alpha gal and its unmasking by desialylation. The means by which alpha gal is sporadically expressed in human tissue remain unknown, as the galactosyl transferase which produces this epitope in constitutively expressive animals has undergone significant mutation at the genomic level in humans. Pathological re-expression is presumed to require permissive changes at a cellular level. Detailing these alterations is a prerequisite to the comprehension of the metastatic phenotype. In this context, the possibility of therapeutic strategies affecting alpha gal expression are also discussed.